

4 – STEP BY STEP SETUP PROCEDURE

When initially setting up a CD30, use the following steps.

NOTE: During initial setup if the controller's on/off signal isn't wired yet or is but isn't being sent, **you must set –En = 1 (under menu C040) or place a jumper between contacts 5 & 6.** This will allow you to operate and run the controller without a run signal coming from a main panel.

4.1 – WIRE CD30/35

Wire all connections to CD30/35, including:

- Incoming plant power
- Outgoing power to the conveyor coil
- The feedback sensor
- Along with any additional control wiring required for the application

NOTE: Reference section 3 for wiring specifics

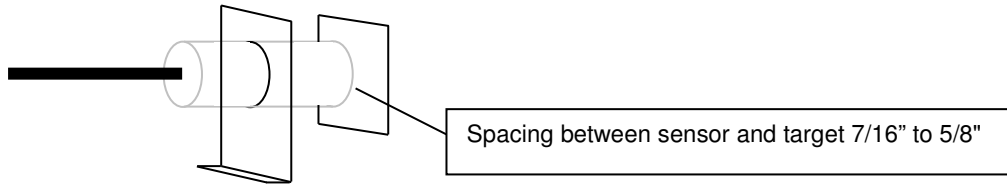
4.2 – INITIAL CD30/35 MENU SETTINGS FOR PPM EQUIPMENT

Note: settings that say “n/a” don't matter for initial setup

Menu C038		Menu C040	
Parameter	Setting	Parameter	Setting
I (current)	100	SIF (interface)	0
A (amplitude)	n/a	I (current)	100
P (power)	n/a	L (low freq. limit)	VFA & Mini = 15 VFP & VFX = 10 Mag II = 25
F (frequency)	n/a		
ESP (External set point)	0		
2. (2nd set point)	n/a	H (high freq. limit)	VFA & Mini = 30 VFP & VFX = 20 Mag II = 35
ACC (loop control)	1		
PA (proportional amplitude gain)	100		
IA (integral amplitude gain)	100	F (actual freq.)	n/a
AFC (auto freq. control)	1	AS (AFC threshold)	25
AFS (start freq. search)		Squiggle up (soft start)	0
		Squiggle down (soft stop)	0.1
		-En (invert enable)	0
		Efu (current regulation)	0
		PI (prop current regulation)	20
		II (Integral current regulation)	40
		rb (status relay)	0
		SPE	0
		APS	1

4.3 – SET SENSOR POSITION

If using the older Gordon sensor (black in color), use menu C051 and adjust the sensor gap until the reading **C051=25**



If using the newer PPM Accelerometer, no adjustment is needed and C051 can be ignored.

4.4 – RUN AUTOMATIC FREQUENCY SEARCH (AFS)

Before normal operation, the controller must perform an AFS to accurately find the resonant frequency of the conveyor spring system. To do this, follow the following steps:

1. Turn the controller on so the conveyor is receiving a power signal
2. Adjust the stroke & power settings to:
 - a. Menu C038, A=0 (sets amplitude to 0)
 - b. Menu C038, P=10 (sets power to 10)
3. Double check the controller is setup to run in the natural frequency control mode by:
 - a. Menu C038, ACC=1 (closed-loop mode)
 - b. Menu C038, AFC=1 (automatic frequency control)
4. Menu C038, AFS, press up arrow (start frequency search)

The readout will show the frequency as the controller searches, when all three lines to the left of the frequency stop blinking the controller has found the resonant frequency and will revert to the normal run root menu.

NOTE: If the controller has a difficult time finding the resonant frequency, the following items may help:

- Try increasing the set point (A) a small amount (by 5 or 10)
- If you know about what the resonant frequency is, try narrowing the frequency range AFS can search (menu C040, parameters L & H)
- Reduce power limit (P)

4.5 – SET OPERATING CHARACTERISTICS (A & P)

For the feedback loop to operate correctly it must be set up for the vibratory feeder that it is used with because every feeder has different response characteristics. Therefore, it is important to adjust the output signal so that amplitude (A) setting of 100% corresponds to the required maximum amplitude of the feeder. To do this, use the following steps:

1. In menu C038 set the (P) value to 10%, and the (A) value to 100%
2. Enable the controller so that the feeder is vibrating
3. Gradually increase the value of (P) until the required maximum amplitude of the feeder is reached

NOTE: A “V” sticker on the side of the feeder tray can be used to give a visual guide if another measuring system is not available.

4.6 – SAVE USER SETTINGS & HIDE MENUS

After setting up the controller, verify that it works to the customer's satisfaction and that all system controllers are working properly perform the following items:

1. Save User Settings (C143) – Go to C143 and push the up arrow to save all settings under the user settings memory
2. Hide Function Menu (C117) – Go to C117 and set HdC=1 to hide the C038 function menu
3. Hide Service Menu (C127) – Go to C127 and set EnS=0 to hide the C040 service menu

NOTE: With the above menus hidden, all the operator can do is adjust the amplitude (A) from the root menu. If the customer more operator adjustment, the above menus can be left unhidden. When they are hidden though there will be less chance of an untrained operator causing a problem with the controller.